Daily IV surface completion with visible same-date anchors

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NQFO Finals

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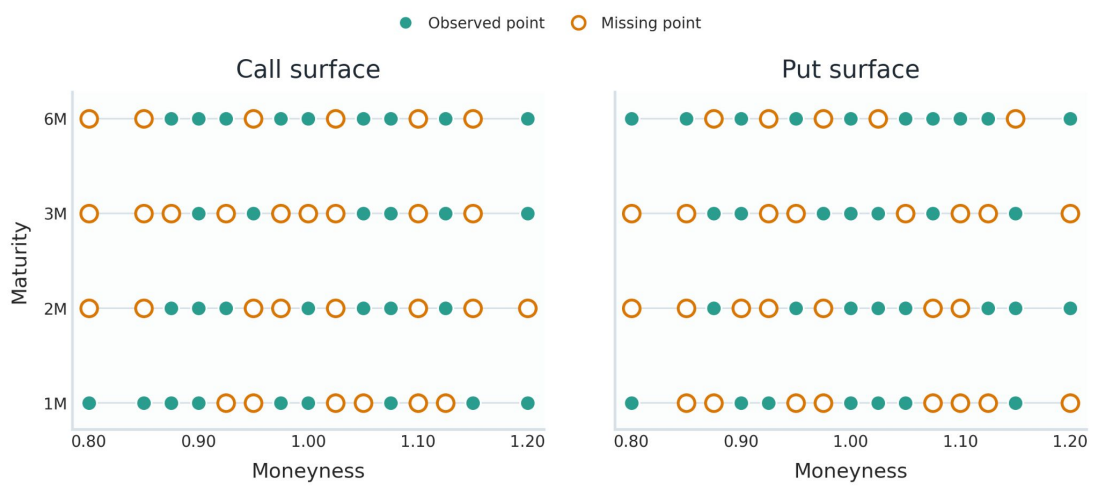
Narrative arc

Task and validation first, then modeling logic, then the locked final hybrid and its remaining weaknesses.

Presentation roadmap

Daily IV surface completion with visible same-date anchors

Each date is a partially observed call–put volatility lattice.



Mathematical setup

$$\mathcal{G}_d = \{(m, T, o)\}, \qquad A_d \subset \mathcal{G}_d, \qquad M_d = \mathcal{G}_d \setminus A_d$$

Estimate $\sigma_d(m, T, o)$ on M_d

using A_d and historical surfaces

d: date, m: moneyness, T: maturity, o: option type

Each date is partially observed; the model must infer the rest of the surface coherently.

Why a complete volatility surface matters

Market quotes are sparse, but pricing and risk systems still need a full surface.

Observed market quotes

Only some strikes and maturities trade actively each day, so many surface nodes are missing.

sparse observations

Complete IV surface

A coherent surface fills the gaps across moneyness and maturity and becomes the market input layer.

downstream use

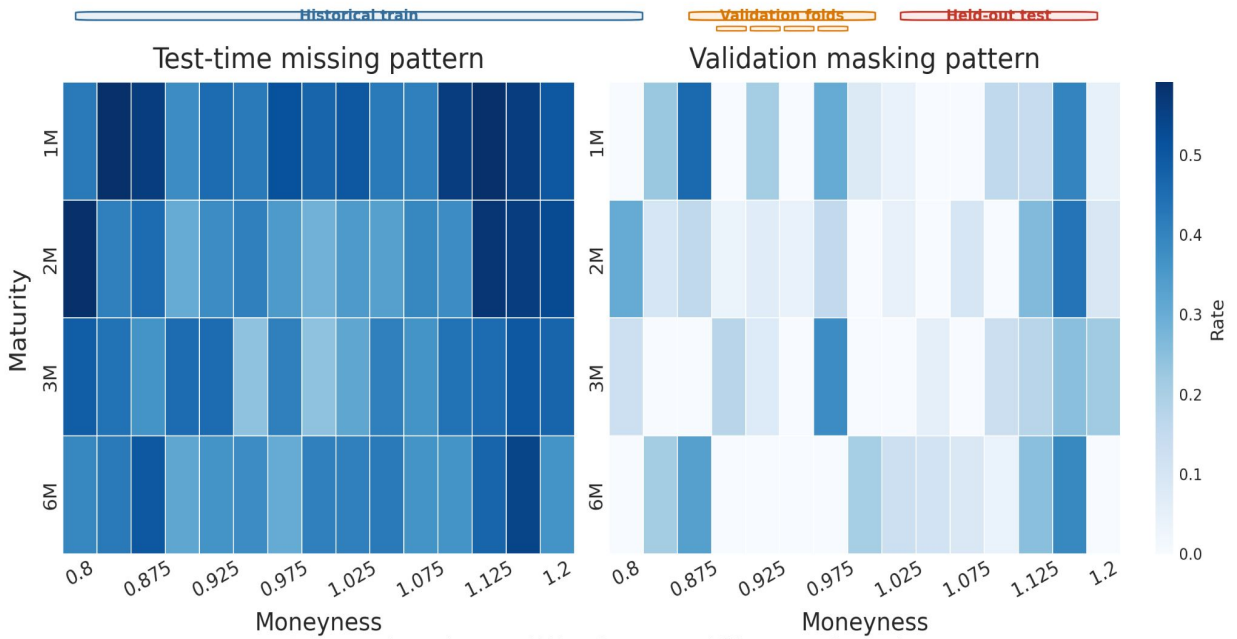
Used for

Pricing
Risk and greeks
Scenario analysis and hedging

Markets give only partial quotes, analysts and option traders still need a complete surface.

Validation design matched the real inference setting

Future dates were predicted using past dates and the still-visible points on the same date.



Only some observed rows are hidden; the rest stay visible as same-date anchors.

Masking logic

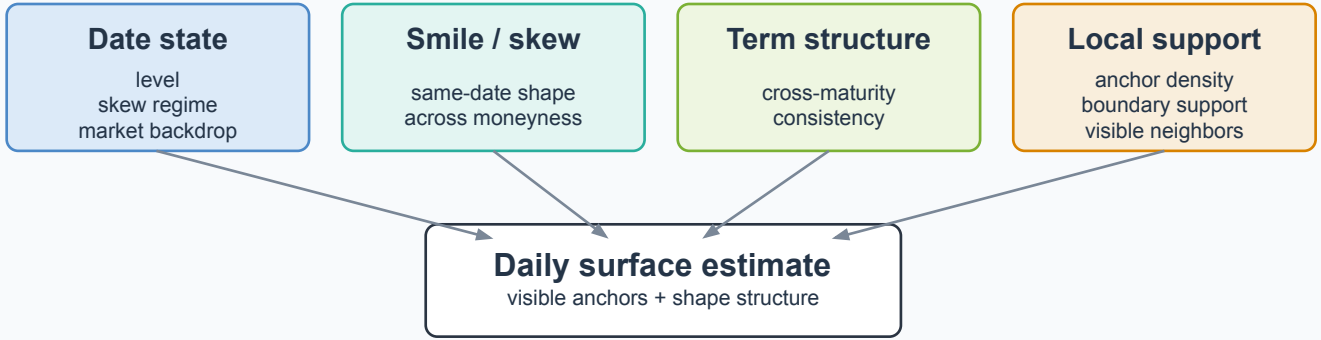
- **Time-ordered split**
  Use earlier dates as history and hold out later dates for validation.
- **Pseudo-hide observed rows**
  On each target date, hide a subset of originally observed rows and score only those hidden rows.
- **Keep same-date anchors visible**
  The remaining observed rows on that date stay visible, just like inference time.

Two validation views were kept: a practical main protocol and a harsher stress protocol.

Validation must preserve both time ordering and the visible same-date anchor points.

Different models capture different signal

Treated model building as a search over structural assumptions about where signal lives in the surface.



One useful modeling decomposition

IV surface signal ≈ date state + smile/skew + term structure + local support

$$\sigma_d(m, T, o) \approx f_{\text{date}}(d) + f_{\text{smile}}(m \mid d) + f_{\text{term}}(T \mid d) + f_{\text{local}}(m, T, o \mid A_d)$$

- Some signal lives at the date level.
- Some lives in cross-sectional shape: smile/skew and term structure.
- Some depends on local support: which nearby points are visible.

The surface combines market state, cross-sectional geometry, and local support — so different model families help in different ways.

From signal sources to model families

Different model families are useful because they encode different structural assumptions about where signal lives.

Signal source	Potential model family(s)	What that family is good at
Date state / market regime	Feature-based regression Latent-factor / low-rank models	Daily level, skew regime, and shared date effects across the surface
Smile / skew across moneyness	Structured surface fitting	Same-date cross-sectional shape along the smile / skew
Term structure across maturities	Structured fitting over total variance Latent-factor / low-rank models	Cross-maturity consistency and shared surface structure
Local support / boundary geometry	Tree-based residual models Graph-style smoothers Targeted correction layers	Nearby visible points, support density, and edge-of-support information

Different model families matter because they impose different beliefs about where useful structure lives in the surface.

Structured backbone: build the first-pass surface

Recover a coherent finance-informed prior from the visible same-date anchors before any ML correction.

Build shape first, then let ML learn only the remaining error.

Visible same-date anchors + history
starting information

→

1. Local interpolation
fill nearby missing nodes

→

2. Smile fit across moneyness
recover same-date shape

→

3. Maturity fit in total variance
fit term structure in $w(T) = \sigma^2(T)T$

→

4. Soft call–put shrinkage
nudge toward opposite option type when visible

→

Structured estimate
first-pass surface

Key idea The structured branch is the surface prior: it uses visible same-date geometry and finance-aware constraints before any residual correction.

Finance-informed surface prior before any ML correction

Final hybrid model: keep structured by default, override selectively

Start from the structured estimate; use specialist overrides only in specific cases.

Router decides whether to keep the backbone or override it.

Default backbone

Structured estimate
finance-informed first-pass surface
default prediction

Optional specialist overrides

Ridge override
small linear specialist

HistGB residual override
nonlinear correction specialist

These are not averaged blindly — they are used only when the router says they help.

Routing decision

Fixed router
validation-selected case rule

center / stable-shape
keep structured

maturity-only subset
use Ridge override

otherwise
use HistGB residual

Locked submission output

Final submitted IV
locked output

Structured backbone + selective specialist overrides; based on validation, no blind averaging

Weakness map

Where the submitted hybrid still struggles on held-out validation — and where the next gains likely are.

Held-out RMSE of the submitted model

The remaining gap is concentrated in wing cases under both validation views.

Weakness -> next step

Weakness	Next step
Wing RMSE remains materially above center under both validation protocols.	Add wing-specific residual modeling and stronger skew-aware inductive bias.
The same pattern persists even in the harsher stress view, so the weakness is structural rather than incidental.	Move from fixed case rules toward confidence-aware routing or learned gating.
The center is already relatively stable, so broad global tuning likely has diminishing returns.	Invest in richer date-level surface state to share information better on difficult dates.

“All models are wrong, but some are useful.” — George Box

Thank you

Thank you

Questions and discussion

Structured prior → Selective overrides → Useful model

Thank you

Appendix: ML feature families used by the two branches

Ridge predicts IV directly; final HistGB predicts the residual to the structured estimate and uses a pruned feature set.

How the two ML branches use context

Ridge branch
predict IV directly

HistGB branch
predict residual and add it back to S

Key difference
prunes raw anchor-value and date-regime families

Feature family	Examples	Why it helps	Ridge	Final HistGB
Surface location	moneyness, log-moneyness, maturity, tau, option type, center vs wing	Tells the model where the row sits on the IV lattice before any correction.	✓	✓
Structured estimates + gaps	same-date interpolation, smile fit, TV maturity fit, structured winner, prediction gaps	Lets ML learn when the structured prior is already reliable and when it needs correction.	✓	✓
Same-date support geometry	visible-neighbor counts, opposite option visible, local support score, hard-case flag	Support quality tells the model whether same-date structure is well anchored or fragile.	✓	✓
Raw anchor values / distances	left-right anchor IVs, anchor distances, local same-date anchor values	Provides direct local context when nearby quotes exist, but is noisier and more mask-dependent.	✓	dropped
Historical node priors	recent-node median, full-node median	Stabilizes sparse rows by borrowing from the same surface node on earlier dates.	✓	✓
Date-level regime proxies	ATM IV proxy, skew proxy, term-slope proxy, anchor ratio, IV dispersion	Captures day-wide level and shape shifts that affect many rows at once.	✓	dropped

Takeaway: Ridge uses the full feature table; the final HistGB keeps row-local, structured, support, and historical-prior signals but prunes raw anchor-value and date-regime families.

Features explain where the row is, how strong the visible support is, and whether the structured prior should be trusted or corrected.

Appendix: prediction cases under the final router

Each row gets three candidate predictions; the fixed router submits exactly one of them.

Shared notation: S = structured estimate | R = Ridge direct-IV prediction | H = HistGB residual prediction = S + Δ

Keep structured
center rows or quadratic-smile-only rows

Use Ridge override
tv_maturity_only subset

Use HistGB residual
all remaining rows

Case A | structured kept

Trigger: center row (or quadratic_smile_only)

Structured S	17.54%
Ridge R	18.01%
HistGB H	17.42%
Selected route	**structured_winner**
Final submitted IV	**17.54%**

ML branches are still computed, but the router keeps the structured prior.

Case B | Ridge override

Trigger: source = tv_maturity_only

Structured S	24.80%
Ridge R	24.10%
HistGB H	25.30%
Selected route	**ridge_direct_full**
Final submitted IV	**24.10%**

This is the narrow conservative override slice retained in the final model.

Case C | HistGB override

Trigger: wing row, not center/quadratic, not tv-only

Structured S	29.20%
Ridge R	29.00%
HistGB H	30.00% = 29.20% + 0.80%
Selected route	**histgb_pruned**
Final submitted IV	**30.00%**

HistGB learns the residual correction; final prediction is S plus the learned delta.

Representative examples (illustrative): compute S, R, and H for each row, then the fixed router selects exactly one final submitted IV.